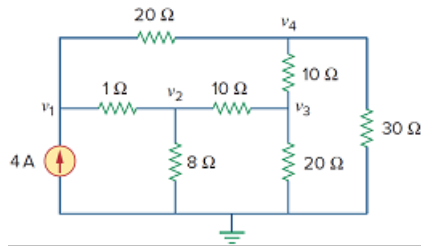


Problem

Use nodal analysis along with *MATLAB* to determine the node voltages in Fig. 3.74.

Figure 3.74:



Step-by-step solution

Step 1 of 7

Consider the following circuit diagram:

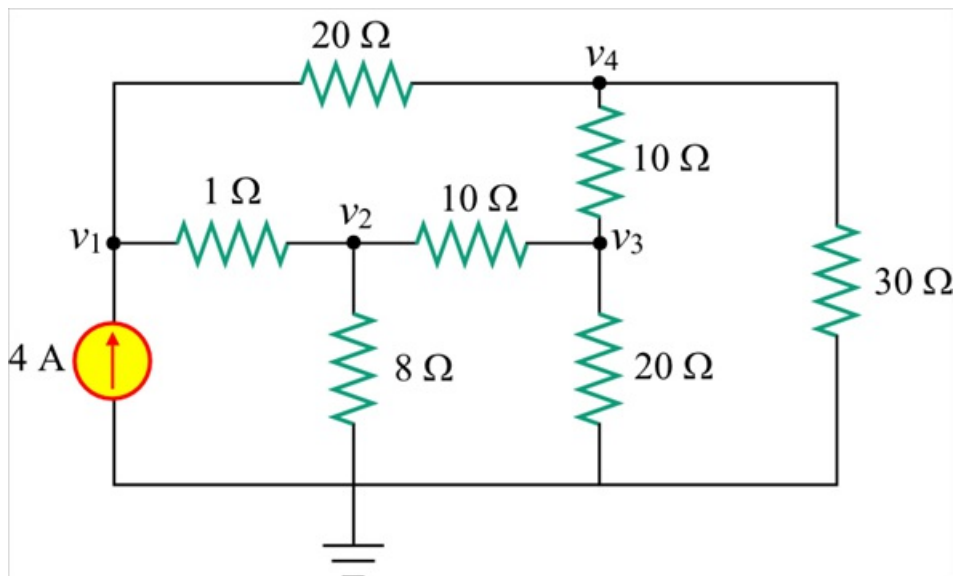


Figure 1

Step 2 of 7

Apply Kirchhoff's current law at node 1.

$$\frac{v_1 - v_2}{1} + \frac{v_1 - v_4}{20} = 4$$

$$\left(1 + \frac{1}{20}\right)v_1 - v_2 - \frac{v_4}{20} = 4$$

$$\left(\frac{20+1}{20}\right)v_1 - \frac{20v_2}{20} - \frac{v_4}{20} = 4$$

$$21v_1 - 20v_2 - v_4 = (4)(20)$$

$$21v_1 - 20v_2 - v_4 = 80 \quad \dots\dots (1)$$

Step 3 of 7

Apply Kirchhoff's current law at node 2.

$$\begin{aligned}\frac{v_2 - v_1}{1} + \frac{v_2}{8} + \frac{v_2 - v_3}{10} &= 0 \\ -v_1 + \left(1 + \frac{1}{8} + \frac{1}{10}\right)v_2 - \frac{v_3}{10} &= 0 \\ -\frac{40v_1}{40} + \left(\frac{40+5+4}{40}\right)v_2 - \frac{4v_3}{40} &= 0 \\ -40v_1 + 49v_2 - 4v_3 &= 0 \\ 40v_1 - 49v_2 + 4v_3 &= 0 \dots\dots (2)\end{aligned}$$

Step 4 of 7

Apply Kirchhoff's current law at node 3.

$$\begin{aligned}\frac{v_3}{20} + \frac{v_3 - v_2}{10} + \frac{v_3 - v_4}{10} &= 0 \\ -\frac{v_2}{10} + \left(\frac{1}{20} + \frac{1}{10} + \frac{1}{10}\right)v_3 - \frac{v_4}{10} &= 0 \\ -\frac{2v_2}{20} + \left(\frac{1+2+2}{20}\right)v_3 - \frac{2v_4}{20} &= 0 \\ -2v_2 + 5v_3 - 2v_4 &= 0 \\ 2v_2 - 5v_3 + 2v_4 &= 0 \dots\dots (3)\end{aligned}$$

Step 5 of 7

Apply Kirchhoff's current law at node 4.

$$\begin{aligned}\frac{v_4}{30} + \frac{v_4 - v_1}{20} + \frac{v_4 - v_3}{10} &= 0 \\ -\frac{v_1}{20} - \frac{v_3}{10} + \left(\frac{1}{30} + \frac{1}{20} + \frac{1}{10}\right)v_4 &= 0 \\ -\frac{3v_1}{60} - \frac{6v_3}{60} + \left(\frac{2+3+6}{60}\right)v_4 &= 0 \\ -3v_1 - 6v_3 + 11v_4 &= 0 \\ 3v_1 + 6v_3 - 11v_4 &= 0 \dots\dots (4)\end{aligned}$$

Write equations (1), (2), (3) and (4) in Matrix form.

$$\begin{bmatrix} 21 & -20 & 0 & -1 \\ 40 & -49 & 4 & 0 \\ 0 & 2 & -5 & 2 \\ 3 & 0 & 6 & -11 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} 80 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Step 6 of 7

The following is the MATLAB code for calculating node voltages.

```
>> a=[21 -20 0 -1;40 -49 4 0;0 2 -5 2;3 0 6 -11];>> b=[80;0;0;0]; >> c=inv(a)*b
```

MATLAB output:

```
c =25.524722.048014.842015.0569
```

Step 7 of 7

The node voltages are,

$$v_1 = 25.52 \text{ V}$$

$$v_2 = 22.04 \text{ V}$$

$$v_3 = 14.84 \text{ V}$$

$$v_4 = 15.05 \text{ V}$$

Therefore, the node voltages, v_1, v_2, v_3, v_4 are 25.52 V, 22.04 V, 14.84 V 15.05 V respectively.